
Test Execution Report

IBA Campus Facility Booking System — Cycle 1 Summary

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Contents

1	Executive Summary	3
1.1	Overview	3
1.2	Summary of Test Run Performance	3
1.3	Release Recommendation	3
2	High-Level Execution Metrics	3
3	Detailed Test Case Status	4
3.1	Authentication & Authorisation (TC-AUTH)	4
3.2	Room Booking Request (TC-BOOK)	4
3.3	Booking Management — Program Office (TC-PO)	4
3.4	Booking Cancellation (TC-CANCEL)	6
3.5	Admin System Management (TC-ADMIN)	6
3.6	Data Integrity & Audit (TC-DATA)	6
4	Failure Analysis and Technical Insights	7
4.1	Unconditional DB Unique Constraint Blockers (DEF-002)	7
4.2	Hardcoded State Mismatch on Cancellation (DEF-001)	7
4.3	Missing DTO Range Validations (DEF-010)	7
4.4	Audit Trail Omissions (DEF-015)	7

1 Executive Summary

1.1 Overview

This document serves as the official Test Execution Report for the first execution cycle of the IBA Campus Facility Booking System. The test suite was executed against the local testing environment consisting of a React frontend, NestJS backend, and an isolated Supabase PostgreSQL instance. Testing covered three distinct layers: unit testing of core services, integration testing of REST API endpoints, and end-to-end (E2E) automated browser workflows (via Playwright).

1.2 Summary of Test Run Performance

The overall test execution yielded mixed results, demonstrating high stability at the unit level but uncovering critical blocker defects during integration and end-to-end testing phases:

- **Total Test Cases Run:** 124 (39 Unit, 43 Integration, 42 E2E)
- **Passed Tests:** 108 (39 Unit, 34 Integration, 35 E2E)
- **Failed Tests:** 16 (0 Unit, 9 Integration, 7 E2E)
- **Overall Pass Rate:** 87.09%

1.3 Release Recommendation

Based on the exit criteria defined in Section 9.2 of the Master Test Plan, the system is **not yet eligible for release**. The exit criteria require a minimum of a 95% test pass rate and a 100% resolution of Severity-1 (Critical) defects. This cycle has left four critical Severity-1 defects active, alongside several Severity-2 issues, primarily involving the database architecture and validation constraints.

2 High-Level Execution Metrics

The table below presents the quantitative pass/fail metrics aggregated by functional module across all test levels.

Table 1: Module-Wise Test Execution Summary

Functional Module	Tests Executed	Passed	Failed	Pass Rate (%)	Status
Authentication & Auth.	28	28	0	100.0%	PASSED
Room Booking Request	16	12	4	75.0%	FAILED
PO Booking Management	13	11	2	84.6%	FAILED
Booking Cancellation	22	13	9	59.1%	FAILED
Admin System Management	44	44	0	100.0%	PASSED
Data Integrity & Audit	1	0	1	0.0%	FAILED
Total Summary	124	108	16	87.1%	UNRESOLVED

3 Detailed Test Case Status

This section tracks the execution status of individual test cases mapped out in the Test Plan, highlighting the failures and linking them to logged bugs.

3.1 Authentication & Authorisation (TC-AUTH)

This module verified credential checks, session termination, and role-based access control (RBAC).

TC ID	Test Case Description	Level	Status	Failure/Defect Mapping
TC-AUTH-001	Student Login Flow	All	Pass	None
TC-AUTH-002	PO Login Flow	All	Pass	None
TC-AUTH-003	Bad Credentials	All	Pass	None
TC-AUTH-004	Protected URL Redirect	Int/E2E	Pass	None
TC-AUTH-005	Student RBAC Enforcement	Int	Pass	None
TC-AUTH-006	PO RBAC Enforcement	Int	Pass	None
TC-AUTH-007	Logout & Back Button Block	E2E	Pass	None

3.2 Room Booking Request (TC-BOOK)

This module verified the user workflow of making reservations, enforcing validation rules, and temporal limits.

TC ID	Test Case Description	Level	Status	Failure/Defect Mapping	Map-
TC-BOOK-001	Happy Path Booking Submission	All	Pass	None	
TC-BOOK-002	Duplicate Slot Rejection	Int/E2E	Pass	None	
TC-BOOK-003	Blocked Slot Rejection	Int	Pass	None	
TC-BOOK-004/7	Boundary: Out-of-bounds slots	Int	Fail	DEF-010: Database crashes (500) due to missing range DTO validation.	
TC-BOOK-008	Missing purpose field check	All	Pass	None	
TC-BOOK-009	Concurrency: Overlapping requests	Int	Fail	DEF-002: Fails due to unconditional unique DB constraint.	
TC-BOOK-010	Past Date Reservation Block	Int	Fail	DEF-003: API accepts historical dates as valid bookings.	

3.3 Booking Management — Program Office (TC-PO)

This module validated PO dashboard listing, approval, rejection, and conflict-resolution processes.

TC ID	Test Case Description	Level	Status	Failure/Defect Mapping
TC-PO-001	View pending requests	Int/E2E	Pass	None
TC-PO-002	PO approves request	Int/E2E	Pass	None
TC-PO-003	PO rejects request	Int/E2E	Pass	None
TC-PO-004	Conflict Auto-Rejection	E2E	Fail	DEF-002: Blocked because second student cannot submit request.

3.4 Booking Cancellation (TC-CANCEL)

This module verified the ability to void bookings, verifying that the canceled slots are returned to the pool.

TC ID	Test Case Description	Level	Status	Failure/Defect Mapping
TC-CANCEL-001	Student cancels pending booking	Int/E2E	Fail	DEF-001: Status is incorrectly marked as 'rejected' instead of 'cancelled'.
TC-CANCEL-002	Student cancels approved booking	Int/E2E	Fail	DEF-004: No Cancel button is rendered on the UI for approved cards.
TC-CANCEL-003	Cancellation exactly at start time	E2E	Pass	None
TC-CANCEL-004	Cancellation after start time block	E2E	Fail	DEF-007: Allows voiding bookings after start time has elapsed.
TC-CANCEL-005	PO cancels approved booking	Int/E2E	Fail	DEF-001: Fails assertion expecting 'cancelled' status payload.

3.5 Admin System Management (TC-ADMIN)

This module validated building, room, and user CRUD operations as well as administrator blocking.

TC ID	Test Case Description	Level	Status	Failure/Defect Mapping
TC-ADMIN-001	Admin adds a new student	All	Pass	None
TC-ADMIN-002	Duplicate ERP constraint check	All	Pass	None
TC-ADMIN-003	Admin adds a PO member	All	Pass	None
TC-ADMIN-004	Admin adds a building	All	Pass	None
TC-ADMIN-005	Admin adds a room	All	Pass	None
TC-ADMIN-006	Admin blocks/disables a room	All	Pass	None
TC-ADMIN-007	Admin unblocks/re-enables room	All	Pass	None

3.6 Data Integrity & Audit (TC-DATA)

This module focused on ensuring the historical state transitions remain valid inside the database schema.

TC ID	Test Case Description	Level	Status	Failure/Defect Mapping
TC-DATA-001	Maintain accurate audit trail	Int	Fail	DEF-015: Fails because reviewed_by is omitted in API SELECT.

4 Failure Analysis and Technical Insights

During this execution cycle, several distinct failure patterns were captured by the testing harness. This section links those failures directly back to code-level root causes.

4.1 Unconditional DB Unique Constraint Blockers (DEF-002)

The failure of TC-BOOK-009 (Concurrency) and TC-PO-004 (Conflict Resolution) is caused by a structural database flaw. In `supabase-schema.sql`, the unique constraint on the bookings table is declared as:

```
UNIQUE (room_id, date, slot_id)
```

Because this constraint is unconditional, the database blocks any entry sharing a slot, regardless of status. Consequently, two students cannot submit concurrent pending requests for the same room and slot, which prevents the Program Office from ever receiving multiple pending options. Furthermore, once a booking is cancelled or rejected, that slot can never be re-booked without triggering a database violation (yielding an unhandled 500 error).

4.2 Hardcoded State Mismatch on Cancellation (DEF-001)

The tests TC-CANCEL-001 and TC-CANCEL-005 failed because of a mismatch between the expected and received status payload. The test expects a status of "cancelled" upon cancellation, but received "rejected". As verified in `bookings.service.ts`:

```
async cancel(id: string, requesterId: string, requesterRole: string) {  
  ...  
  return this.updateStatus(id, 'rejected', requesterId);  
}
```

By routing cancellations to the rejection workflow, the application overwrites the audit history and returns incorrect metadata to the client.

4.3 Missing DTO Range Validations (DEF-010)

Test case TC-BOOK-004/7 failed because the API endpoints do not gracefully reject out-of-bounds slot IDs (such as `slot_id: 0` or `slot_id: 8`). Instead of returning a 400 Bad Request, the backend throws an unhandled 500 Internal Server Error because the invalid ID directly violates the foreign key constraint on the database. The `CreateBookingDto` in `bookings.service.ts` only checks if the input is an integer but fails to restrict the range to `[1, 7]`.

4.4 Audit Trail Omissions (DEF-015)

During TC-DATA-001, the integration runner fetched a newly approved booking from the API expecting to verify the reviewer's identity. However, the assertion failed because the returned payload was missing the `reviewed_by` attribute entirely. This is due to an explicit omission within the SQL `SELECT` projection string in `bookings.service.ts`, which prevents the audit trail from being displayed on the frontend.